Chapter 6

Feed-water control and instrumentation

6.1 The principles of feed-water control

The objective of a feed-water control system may seem simple: it is to supply enough water to the boiler to match the evaporation rate. But as is so often the case with boilers, this turns out to be a surprisingly complex mission to accomplish. There are difficulties even in making the basic drum-level measurement on which the control system depends. The design of the control system is then further complicated by the many interactions that occur within the boiler system and by the fact that the effects of some of these interactions are greater or smaller at various points in the boiler's load range.

The control-system designer's task is to develop a scheme that provides adequate control under the widest practicable range of operational conditions, and to do so in a manner that is both safe and cost-effective. To do this it is necessary to understand the detailed mechanisms of the feed-water and steam systems and to be fully aware of the operational requirements.

In all but the smallest and simplest boilers, each of the interrelated factors has to be taken into account, and it is insufficient to rely on simple responses to the three parameters which seem to be relevant to the supply of feed water: steam flow, feed-water flow and the level of water in the drum.

6.2 One, two and three-element control

The level of water in the drum provides an immediate indication of the water contained by the boiler. If the mass flow of water into the system is

greater than the mass flow of steam out of it, the level of water in the drum will rise. Conversely, if the steam output is greater than the feed inflow, the level will fall.

As stated in Chapter 2, the purpose of the drum is not only to separate the steam from the water but also to provide a storage reservoir that allows short-term imbalances between feed-water supply and steam production to be handled without risk to the plant. As the level of water in the drum rises, the risk increases of water being carried over into the steam circuits. The results of such 'carry-over' can be catastrophic: cool water impinging on hot pipework will cause extreme and localised stresses in the metal and, conversely, if the level of water falls there is a possibility of the boiler being damaged, partly because of the loss of essential cooling of the furnace water-walls.

Therefore, the target of the feed-water control system is to keep the level of water in the drum at approximately the midpoint of the vessel. Given this objective, it would appear that the simplest solution would appear to be to measure the level of water in the drum and to adjust the delivery of water to keep this at the desired value—feeding more water into the drum if the level is falling, and less if the level is rising. Unfortunately, the level of water is affected by transient changes of the pressure within the drum and the sense in which the level varies is not necessarily related to the sense in which the feed flow must be adjusted. In other words, it is not sufficient to assume that simply because the level is increasing the feed-water flow must be decreased, and vice versa.

This strange situation is due to effects known as 'swell' and 'shrinkage'. Boiling water comprises a turbulent mass of fluid containing many steam bubbles, and as the boiling rate increases the quantity of bubbles that is generated also increases. The mixture of water and bubbles resembles foam, and the volume it occupies is dictated both by the quantity of water and by the amount of the steam bubbles within it. If the pressure within the system is decreased, the saturation temperature is also lowered and the boiling rate therefore increases (because the temperature of the mixture is now higher in relation to the saturation temperature than it was before the pressure change occurred). As the boiling rate increases, the density of the water decreases, but since the mass of steam and water has not changed the decrease in density must be accompanied by an increase in the volume of the mixture.

By this mechanism the level of water in the drum appears to rise, a phenomenon referred to as 'swell'. The rise of level is misleading: it is not indicative of a real increase in the mass of water in the system, which would require the supply of water to be cut back to maintain the status quo. In fact, if the drop in pressure is the result of the steam demand

suddenly increasing, the water supply will need to be increased to match the increased steam flow.

'Shrinkage' is the opposite of swell: it occurs when the pressure rises. The mechanism is exactly the same as that for swell, but in the reverse direction. Shrinkage causes the level of water in the drum to fall when the steam flow decreases, and once again the delivery of water to the boiler must be related to the actual need rather than to the possibly misleading indication provided by the drum-level transmitter.

If a slow change of steam flow occurs, all is well because the pressure within the system can be controlled. It is when rapid steam-flow changes happen that problems occur since, due to swell or shrinkage, the drum-level indication provides a contrary indication of the water demand.

Following a sudden increase in steam demand, which causes the pressure to drop (and therefore the drum level to rise), a simple level controller would respond by reducing the flow of feed water. Equally, a sudden decrease in steam flow, which would be accompanied by a rise in pressure and an attendant fall in the drum level, would cause a level controller to increase the flow of water. Both actions are, of course, in the incorrect sense.

The effects of swell and shrinkage, in addition to being determined by the rate of change of pressure, also depend on the relative size of the drum and the pressure at which it operates. If the volume of the drum is large in relation to the volume of the whole system the effect will be smaller than otherwise. If the system pressure is low the effect will be larger than with a boiler operating at a higher pressure, since the effect of a given pressure change on the density of the water will be greater in the low-pressure boiler than it would if the same pressure change were to occur in a boiler operating at a higher pressure.

Faced with this situation, designers of control systems have responded by implementing a variety of solutions. The simplest of these is a 'two-element' system, since it is based on the use of two process measurements in place of the single drum-level measurement used above.

6.2.1 Two-element feed-water control

Remembering that the basic requirement of a feed-water control system is to maintain a constant quantity of water in the boiler, it is apparent that one way of addressing the problem would be to maintain the flow of water *into* the system at a value which matches the flow of steam *out* of it. One version of this system is shown in Figure 6.1. Here, the flow is controlled by an easily recognised device, a valve. We shall look at valves in more depth later, but for the moment assume that the version used in the diagram

maintains the rate of water flowing through the valve at a figure which is directly proportional to the demand signal from the controller (i.e. if the demand signal varies linearly from 0 to 100%, the flow rate also changes linearly between 0 and 100%). Such a valve is said to have a 'linear characteristic' and in the system shown this is employed in conjunction with a transmitter that produces a signal proportional to steam flow. Used together, these two devices keep the parameters in step. If the transmitter produces a signal which is equal to the steam flow at all loads and if the flow through the valve is matched with this signal at every point in the flow range, a controller gain of unity will ensure that, throughout the dynamic range of the system, the flow of water will always be equal to the flow of steam.

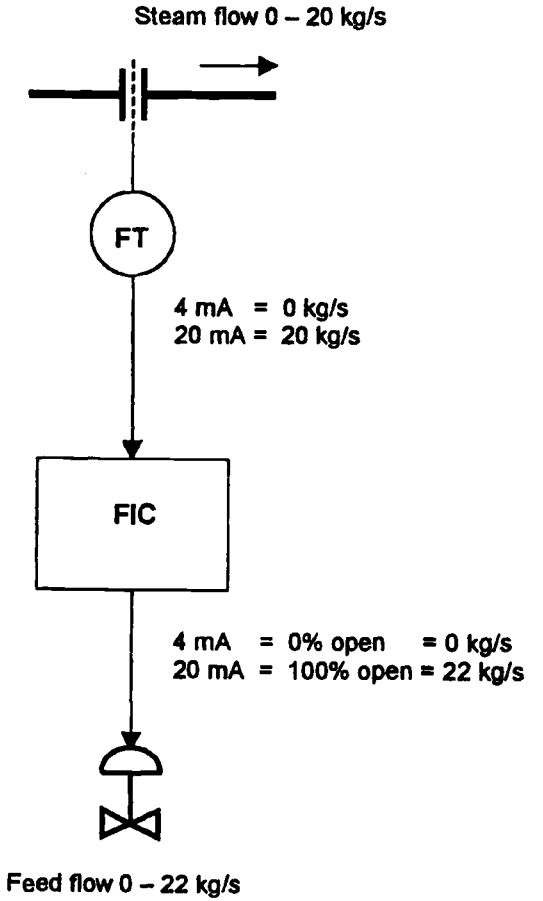


Figure 6.1 Feed-water control based on feed-flow measurement only

Naturally, scaling factors of the transmitter and the valve must be taken into account. If the range of the flow transmitter is different from the valve's flow-control range, the controller gain will need to be adjusted accordingly, and in practical systems this is always necessary.

In order to provide an adequate operational margin of confidence, the range of the control valve is always designed to be greater than the flow range of the boiler. For example, in a boiler producing 20 kg/s of steam, the valve may be sized to deliver 22 kg/s of water when it is fully open. In this example, with a linear valve characteristic, an opening of approximately 91% will be needed to pass a flow of 20 kg/s.

In this case, if the steam-flow transmitter produces an output of 100% at 20 kg/s flow, the controller gain must be such that a measured value of 100% produces an output of 91%. This is a proportional band of 110 (i.e. a gain of 20/22) and if this gain is assigned to the controller the feed flow will match the steam flow over the entire range of boiler load (assuming that the valve characteristic is linear, that the flow transmitter output is 4 mA at zero flow, and that zero flow of water occurs with a valve signal of 0%).

The problem with this system is that it only matches the steam- and feed-flow rates. If, at the outset, the drum level is below the desired value, that is where it will stay, because if everything is set up correctly the feed into the boiler will always match the steam flowing out of it, and there is no mechanism for introducing the small surfeit of feed over steam, or the slight deficit, that is needed to correct the drum-level error.

It is important to consider the practical reality of what would happen if things were *not* to be set up correctly. In this situation, if there is a small setting error in the controller gain, or if the feed valve passes more or less water than it should at the given opening, or if the steam flow transmitter is slightly out of calibration, the drum level will integrate up or down at a rate determined by the scale of the error, and nothing will correct for this undesirable state of affairs.

In other words, the system cannot correct the drum level if this parameter deviates from the desired value either because of an initial error or because of small errors in the steam-flow measurement or nonlinearities between the valve demand and the actual flow through it. In the example given above, the exact gain required is 0.909 09.... Therefore, if the controller gain were to be set to 0.91 as suggested above, the feed-water flow would be slightly greater than the steam flow, and the drum level will gradually increase.

To counter these effects it is necessary to add a feedback element, consisting of another controller which will act to correct for any mismatch

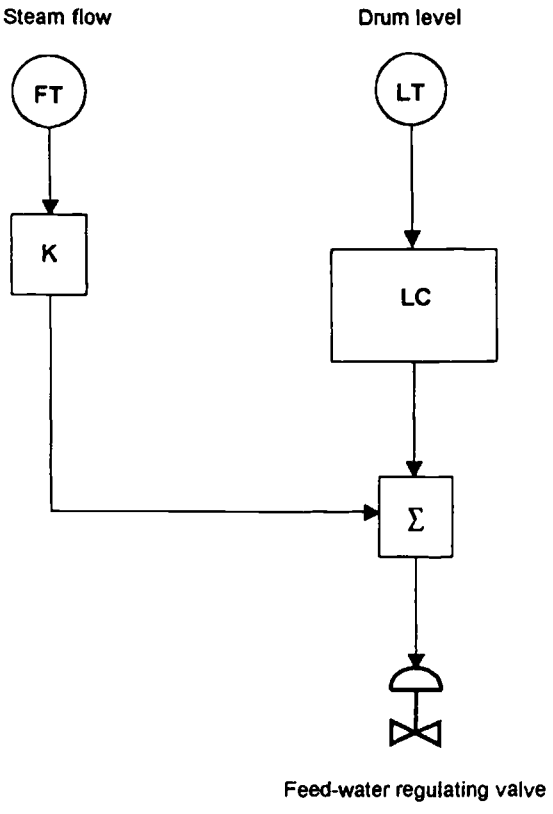


Figure 6.2 Basic two-element feed-water control system

between the actual and desired drum levels. Figure 6.2 shows one variety of such a 'two-element' system.

In such a system, because the drum volume and the steam and feed flows form an integrating system, with the drum level integrating any steam-flow/feed-flow mismatch, it is unnecessary to employ an additional integration function in the controller. Therefore the drum-level controller should be of the proportional-only variety.

The correct gain for this controller can be determined from a knowledge of the swell and shrinkage effects within the boiler. If these are not known they can be determined by test. A suitable test would be to change the steam flow as rapidly as possible by, say, 10% of the maximum evaporation rate of the boiler, while keeping the feed flow in step with the steam flow. (This can be achieved by hooking a feed-flow signal into the system while temporarily disabling the drum-level controller).

To see how the information on the boiler's swell characteristic can be used to help with controller tuning, let us examine a two-element system

where the range of the steam-flow transmitter is ranged as above (0–20 kg/s), and the feed valve is again sized to deliver 22 kg/s when it is 100% open. Assume that the drum-level transmitter is ranged to produce 4 mA when the water level has dropped below the setpoint by 250 mm, and that it is 20 mA when the level is 250 mm above the setpoint (i.e. a range of 500 mm). Finally, assume that a test as described above has determined that the swell resulting from a sudden 10% change of steam flow raises the drum level by 80 mm.

If the drum-level controller is to exactly counteract the effect of swell, it must produce an output that cancels out the step change in the steam flow, which was 10%. The controller output must therefore change by 10% when the input error changes by 16%, which means that the gain must be 0.625 ($10 \div 16$).

When the steam-flow and pressure changes have settled out and the water level has returned to the setpoint, the level-controller output will again become zero. The valve opening will then revert to tracking slow changes in the steam flow, as described earlier.

This analysis depends on the swell effect being constant over the boiler load range, which may or may not be true, but it provides a practical method of tuning this type of system, and will produce a fairly good performance over a wide range of conditions. Theoretically, better results could be obtained by carrying out tests to determine the swell effect at various points in the load range and introducing a nonlinear function within the level controller to compensate for the differences across the range. But this is rather complicated for what is essentially a simple system and in any case performance is likely to be limited by the other serious deficiencies within the system, which we shall examine in the next section, which discusses a more comprehensive system, known as three-element control.

6.2.2 Three-element feed-water control

Throughout the above analysis, reference has been made to the feed-water valve characteristic being linear and the valve being sized to produce a fixed flow when it is 100% open. However, the flow through a valve depends both on its opening and on the pressure drop across it. In a feed-water system, the pressure drop across the valve varies from instant to instant, and the flow through it at any given opening will therefore vary. For reasons given earlier, in a simple two-element system based on drum level, the inclusion of an integration element in the level controller is undesirable. Therefore the varying flow results in the level control becoming offset, to restore the steam-flow/feed-flow balance. This offset is undesir-

able, since it needlessly erodes the safety margin provided by the presence of the drum.

One method of correcting for the error produced by the feed valve is the addition of a third element to the system—a measurement of feed-water flow.

There are various ways of implementing such a system, one of which is shown in Figure 6.3. Here, the output of the drum-level controller is trimmed by a signal representing the difference between the feed-flow and steam-flow signals. A gain block (4) is introduced to compensate for any difference between the ranges of the two transmitters. In most cases the steam-flow and feed-flow signals will cancel out, and the drum-level controller will be modulating the feed flow to keep the level at the setpoint. In this case, it is reasonable to apply an integral term in this controller, as shown.

In another implementation of this familiar system shown in Figure 6.4, a 'cascade control' technique is applied. The drum-level controller (item 5,

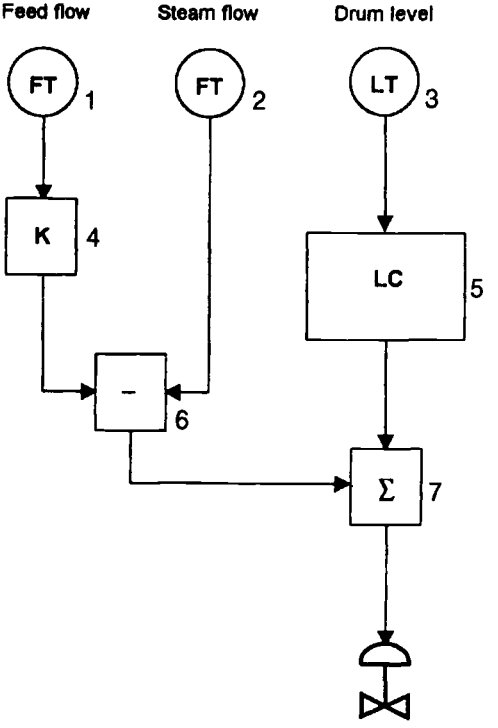


Figure 6.3 One type of three-element feed-water control system

proportional only), compares the measured level signal with a set value and produces a bipolar output proportional to any error. This trims a modified steam-flow signal, which is acting as the desired value for a closed-loop feed-water controller (7). As previously, a gain block (4) adjusts for any range difference between the steam-flow and feed-flow transmitters.

These are not the only ways of implementing three-element control. Several variants of the system are in common use, each with its own advantages and disadvantages. However, each system has one factor in common, the use of steam-flow, feed-flow and drum-level measurements. The application of the feed-flow measuring element compensates for any variations in feed-flow, whether these are due to the pump characteristics or other factors, and the three-element system is therefore recommended wherever accuracy of control is required.

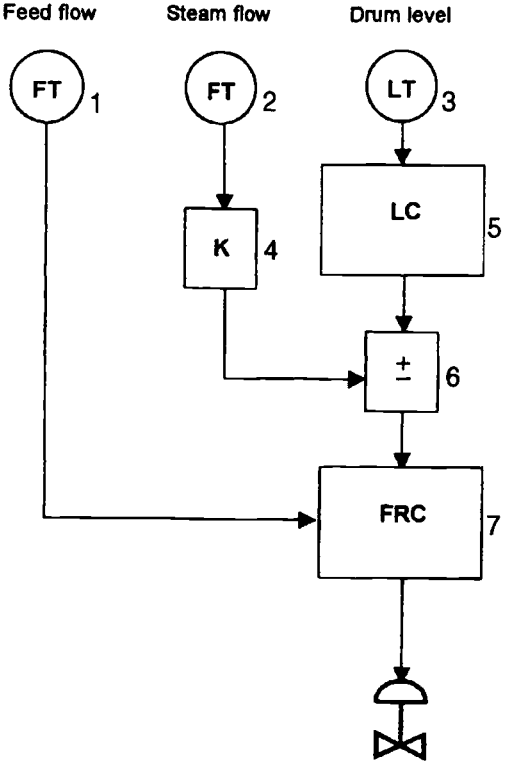


Figure 6.4 Alternative (cascade) three-element feed-water control system

As might be expected, a three-element system is more expensive than a single-element or two-element system, but it is always worthwhile to examine the real cost differences before deciding on which system to use. For example, the use of a steam-flow signal may not, in fact, add any additional cost, because this measurement may be needed elsewhere, e.g. in the combustion control system.

However, the presence of a feed-flow measurement in the system does necessarily add a significant cost burden. The transmitter itself is a complex and not inexpensive item and it may also require the provision of a primary element (such as a Venturi, flow nozzle or orifice plate) which will add further cost and require the provision of adequate lengths of straight pipework upstream and downstream of the device.

The capital cost of the system needs to be carefully considered against the background of a detailed knowledge of the operational régime of the plant. All too often, insufficient rational thought is applied to these factors. The use of a small drum in the boiler offers a not insignificant reduction in the cost of a plant, but its use also reduces the operational safety margins and therefore requires increased accuracy of feed-water control. Yet, the cost-cutting exercises that lead to the use of a small drum are very often extended to the control system as well, further reducing the margin for error. If safety margins are not to be dangerously eroded, careful consideration must be applied to the determination of practical design rules for feed-water control systems. The control implications should be given just as much consideration as the process or mechanical aspects.

6.3 Measuring and displaying the drum level

From the above, it is apparent that the primary objective of the feed-water control system of a drum-type boiler or HRSG is to maintain the drum level at the correct value. We shall now look at how this parameter is measured. It is an area where the problems are unexpectedly complex.

Figure 6.5 shows one method of measuring the drum level. This connects the differential-pressure transmitter directly to the drum, via isolating valves. Note the 'constant-head' reservoir connected to the upper tapping point. Because the impulse pipework to the transmitter is outside the heated zone of the boiler, any steam within it will tend to condense, and the pressure applied to the HP port of the transmitter will therefore be the steam and water pressure plus the pressure due to the weight of this condensate.

The latter will depend on the volume of condensate that has collected, and this will be time-dependent. The pipe will be full of steam after the

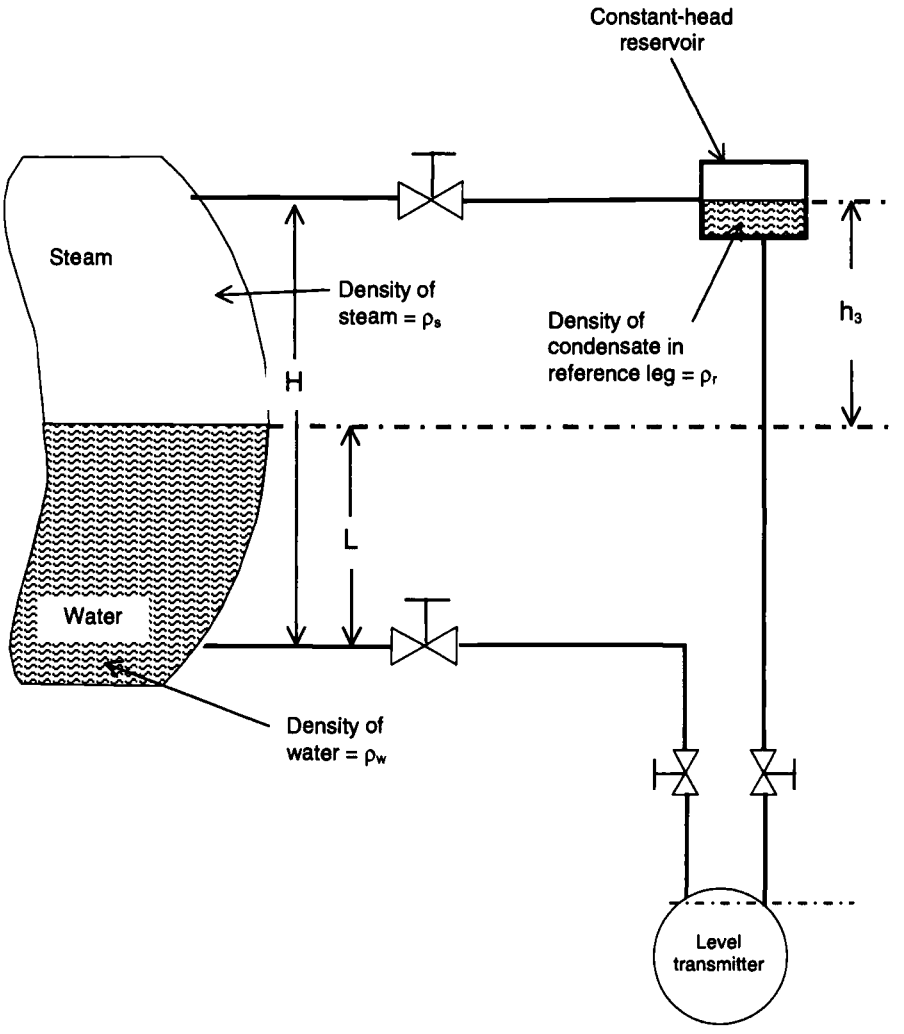


Figure 6.5 Measuring the drum level without a water column

transmitter impulse lines have been 'blown down' on start-up, but it will afterwards start to fill with condensate. The constant-head vessel is left deliberately unlagged, so that the steam in it condenses, maintaining the pipe to the transmitter full of condensate.

The differential-pressure appearing at the transmitter ports is a function of several variables: the water level in the drum, the densities of this water and the steam above it and the density of the water in the pipes to the transmitter. In addition, the derivation of a level signal needs to take

into account the density of the fluid used during the initial calibration of the transmitter. Finally, it is also necessary to recognise the requirement that the lowest drum-water level at which the system will be allowed to operate will be at some point above the level of the lower tapping point.

The differential pressure at the transmitter is defined by*

$$DP = (1/\rho_r) \times [(H \times \rho_c) - [(L \times \rho_w) + (H - L) \times \rho_s)]]$$

where

DP = the differential pressure at the transmitter (at 20 °C)
H = the distance between the bottom and top tappings
L = the height of water in the drum above the bottom tapping
ρ_r = the density of water used for calibrating the transmitter (at 20 °C)
ρ_s = the density of the water in the drum at the operating pressure
ρ_w = the density of the steam in the drum at the operating pressure

It should be noted that the differential pressure at the transmitter will be highest when the separation between the water level and the top tapping is the greatest i.e. at low levels. In other words, a 4–20 mA transmitter will produce 4 mA at the highest level and 20 mA at the lowest. This apparently reversed 'sense' must be corrected by the DCS before the density calculation is applied.

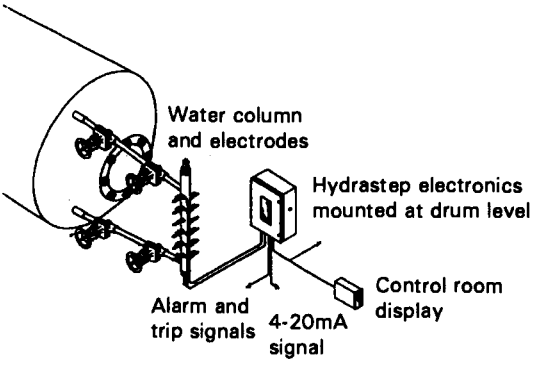
Since the densities of the steam and water in the drum will both depend on the conditions that exist in the drum, the pressure needs to be taken into account when calculating the actual level based on the differential head produced at the transmitter. At one time, various proprietary devices were available for performing the required calculation, but it is most economical, these days, to perform the calculation in the DCS, based on the differential-pressure and pressure signals. In this case the above equation will enable the level to be calculated for any combination of signals from the two transmitters.

However, an important point to bear in mind when using the DCS to perform the pressure compensation in this way is that the corrected signal will be available to the operator *only while the DCS is operational.* If a major failure should occur in the computer system, it is important that the operator can still be able to monitor the drum level by other means, and these must be compensated in a similar way to the above, so that the indication is relatively unaffected by pressure.

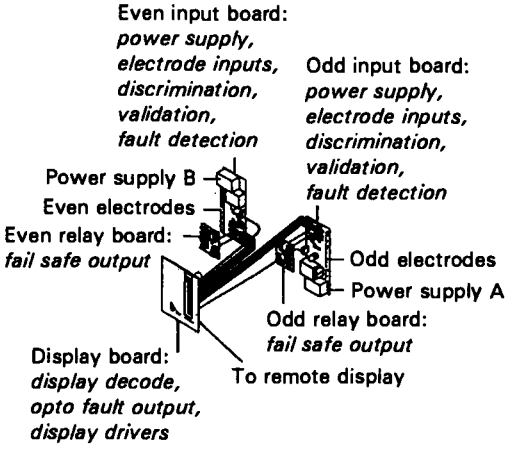
Figure 6.6 shows a proprietary system which generates a drum-level signal for display locally and in the control room. This technology is based on the fact that the conductivity of water is different from that of steam,

* Courtesy of Bristol-Babcock Ltd, Kidderminster, Worcestershire, UK

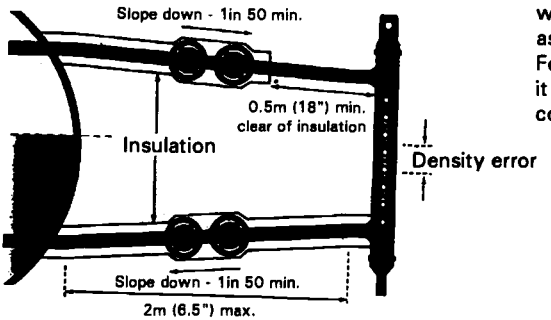
(a)



(b)



(c)



For optimum performance, and to minimize density error, Hydrastep water columns should be installed as shown here.
For columns longer than 1m. (3'3"), it may be necessary to insulate the column itself.

Figure 6.6 The 'Hydrastep' level-indicating system
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and a series of electrodes mounted in a water column attached to the drum uses this fact to detect the interface between water and steam. (The use of an external water column is necessary because of the number of penetrations that would otherwise be necessary in the boiler drum.) The detection circuit is divided into two groups of 'even' and 'odd' electrodes, so that failure of a single drive circuit cannot disable the entire system.

Although such a system will be affected by the difference in the densities of the water in the drum and the column, careful design of the installation will minimise any errors (Figure 6.6*c*).

At present, although such devices provide an excellent indication of the drum level, they are not suitable for control, because of the transient disturbance that occurs as the level moves from the position of one electrode to another. These step changes can produce unpredictable effects in the control loop.

6.3.1 Using an external water column

Although the method of connection shown in Figure 6.5 is viable, it has the disadvantage of being sensitive to errors during sudden reductions in the boiler pressure, caused by the condensate 'flashing off' — boiling as the temperature of the fluid suddenly finds itself above the saturation temperature.

An arrangement that minimises this problem is shown in Figure 6.7, where an external water column is connected to the drum, so that the level of water in column is (theoretically) the same as the level within the drum. The column uses a volume of stored fluid which is larger than the volume of condensate in the small-bore HP leg of Figure 6.5, and the system is therefore less vulnerable to flashing off.

As with the detection column shown in Figure 6.6, great care must be taken to avoid errors caused by the temperature of the fluid in the measuring system being very different from that of the steam and water in the drum. This leads to a density error, since the water column at the gauge will balance with the level in the drum, although its length is less than the distance between the lower tapping and the level of the water within the drum.

Considering the drum and the column of water in the column to be a 'U' tube, it will be seen that balance will occur when the weight of fluid in the left-hand leg equals the weight of fluid in the right-hand leg.

By keeping the temperature conditions within the column as close as possible to those in the drum, the density error will be minimised. This is done by arranging the pipework so that fluid flows through the water column. Unlike the drum itself and all the other pipework, the water

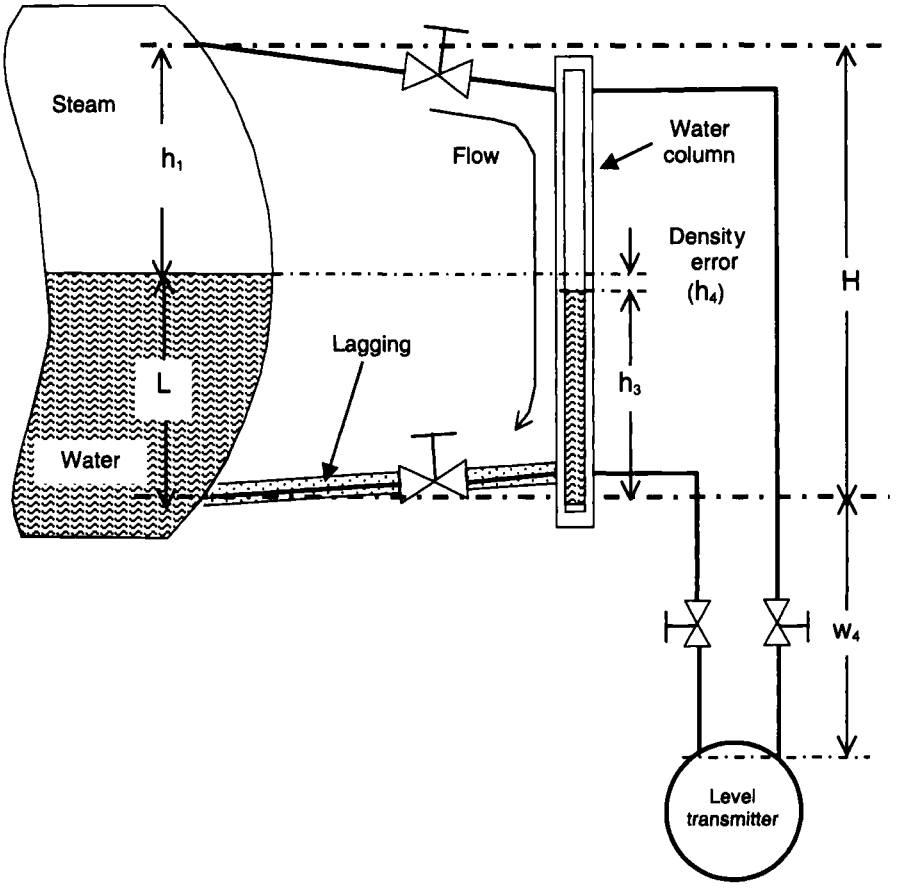


Figure 6.7 Drum-level measurement with a column

column and the section of pipe connecting it to the tapping-point isolation valve (on the right in Figure 6.6) are left unlagged so that the steam condenses and the impulse line remains full of condensate. A circulation of fluid is then established through the column, with steam leaving the drum at the upper tapping point, condensing in the pipe and water column, and returning to the drum via the lower tapping point. This flow tends to maintain the temperature within the water column at a value which is as close as possible to the condition within the drum. Nevertheless, some temperature difference will still exist, and this will have the effect of increasing the density of the fluid in the cooler parts of the system.

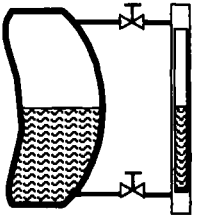
6.3.2 Statutory requirements

In many countries, there is a legal requirement to provide separate systems to make the boiler operator aware of the level of the water in the drum. In the UK, British Standard BS 1113 : 1998* refers to this subject in section 7.3, as follows:

> Each steam boiler, in which a low level, or into which a low flow rate, of water could lead to unsafe conditions shall have at least two independent and suitable means of indicating the water level or flow. Each indicating device shall be capable of being isolated from the boiler and each device shall be a water level gauge in which the water level can be observed except in the following cases:
>
> (a) ... (refers to once-through boilers)
> (b) For boilers with any safety valve set at or above 60 bar *g*, the use of two independent manometric remote water level indications shall be permitted in place of one of the water level gauges. In such cases these remote water level indicators shall have their own independent connections to the boiler.
> (c) For boilers of less than 145 kg/h evaporative capacity, one water level gauge is sufficient.
> (d) The use of alternative devices in place of water level gauges in which the water level can be observed shall be permitted, subject to agreement between the manufacturer and the Inspecting Authority. The design of the devices shall combine appropriate design principles such as fail-safe modes, redundancy, diversity and self-diagnosis in order to provide suitable and reliable indication.
>
> The water level gauge in which the water level can be observed shall be mounted so that the lowest water level that can be observed is at least 50 mm above the lowest water level at which there will be no danger of overheating any part of the boiler, when in operation at that level. Where this is not practicable, the water level gauges shall be sited by agreement with the Inspecting Authority in positions that have been found by experience to indicate satisfactorily that the water content is sufficient for safety under all service conditions.
>
> At least one water level gauge with its isolating valves or cocks shall be connected directly to the boiler, and other than a drain, no device shall be fitted to the gauge that could cause incorrect indication of the water level in the gauge.

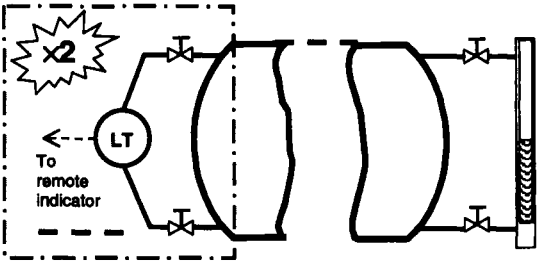
Figure 6.8 is an attempt to illustrate some of the above requirements pictorially. Note that the aim of this diagram is only to illustrate the requirements and to provide general guidance. It is important that a

* Extracts from BS 1113 : 1998 are reproduced with the permission of BSI under licence no. PD\1998 1883. Complete editions of the standards can be obtained from BSI Customer Services, 389 Chiswick High Road, London, W4 4AL, UK.



A 'water level gauge in which the level can be observed ...'.

Note that at least two such gauges are required on boilers of 145 kg/h (0.04 kg/s) or above.



'For boilers with any safety valve set at or above 60 bar *g*, the use of two independent manometric remote water level indications shall be permitted in place of one of the water level gauges. In such cases these remote water level indicators shall have their own independent connections to the boiler.'

'....The use of alternative devices in place of water level gauges in which the water level can be observed shall be permitted, subject to agreement between the manufacturer and the Inspecting Authority. The design of the devices shall combine appropriate design principles such as fail-safe modes, redundancy, diversity and self-diagnosis in order to provide suitable and reliable indication.'

Figure 6.8 Permitted alternatives for water-level measurement on drum-type boilers and HRSGs
Based on BS 1113 : 1998

specific installation is designed to meet the standards required at the actual point of use, since the standards set by authorities in the relevant country, or by the insurers, may differ substantially from those indicated by this diagram.

The above Standard also refers to low-level alarms, stating that

> Every steam boiler shall be provided with a low water level or low water flow audible alarm device.
>
> Water level alarms, whether of low water or high and low water type, shall be so fitted that the alarm is actuated while the level is still visible or indicated in the water level gauges.

With reference to the size of the connecting pipework BS 1113 : 1998 states that:

> Where a water level gauge, safety control or alarm device is connected to the boiler by pipes, the bore of such pipes shall not be less than 25 mm.

(The standard allows the ends of the pipes local to the fittings to be reduced to not less than 20 mm bore for water level gauges and to 25 mm bore for separate safety control and alarm devices.)

In this respect the Standard differs from others, but on one point there is agreement, there is a need to use comparatively large-bore pipework for instrument connections. The use of such pipes increases the cost of the

installation, but their use is important from the viewpoint of safety, since small-bore pipework can become obstructed (for example by sludge) and an obstructed impulse pipe can lead to an instrument providing seriously incorrect information on the parameter being measured. (The use of very small-bore pipes can also cause a reading to be sluggish because of the time taken for pressure changes to affect the instrument, though this is less of a problem with today's instruments which displace very small volumes for full-scale operation.)

6.3.3 Discrepancies between drum-level indications

It sometimes happens that various instruments connected to the same boiler drum display level measurements that are significantly different from each other. Since it is unlikely that the actual drum is anything but horizontal (except for installations on submarines during diving operations) such discrepancies must be due to some error or other. The following list summarises the factors that can cause errors:

- *Density errors*: Differences between installations can cause one instrument to be more affected by density factors than another. (One possibility is that, inadvertently, lagging has been applied to one of the condensation reservoirs).
- *Turbulence*: The surface of the boiling water inside the drum is anything but still. It has been known for 'standing waves' to exist around the downcomers, affecting some measurement points more than others.
- *Flashing-off*: Differences in the geometry of the measuring systems can cause some measurements to be more affected than others by flashing-off during pressure changes.
- *Calibration*: It is vital that all transmitters are carefully and accurately calibrated, and that any density compensation is correctly set up.
- *Installation*: As stated earlier, errors or sluggish response can be the result of partial or complete plugging of impulse lines, or imperfect blow-down operations.

6.3.4 Steam extraction

In both the two-element and three-element systems, an assumption made in the above examples is that the steam output by the boiler eventually returns to the inlet in the form of water. This is not true where significant losses occur or if any steam is abstracted for applications such as soot-blowing. Here the steam is effectively lost, and if a soot-blower abstraction is made at the drum the amount being used will not be included in the flow measurement. This will result in the drum level being offset from the desired value since the flow of feed water into the system should be equal to the steam taken by the load (plant or turbine) plus the steam used for soot-

blowing. The effects of such operational factors will be particularly significant where the design of the control system has been based on the use of a proportional-only controller. In such cases it may be necessary to add a small degree of integral action to the controller, although this should be restrained since the additional integral action can affect the stability of the system.

6.4 The mechanisms used for feed-water control

In this analysis we have looked at the principles of control, and seen that because of various problems, a variety of control methods has evolved, the selection of which depends on a variety of engineering and economic considerations relating to each application. In the discussion some reference had to be made to the mechanism for controlling the flow and for simplicity it was assumed that this was by means of a familiar device, a valve. Now we shall look at the nature of valves in greater depth, and then we shall examine other methods of controlling the flow.

6.4.1 Valves

What follows is merely a practical overview of valve designs in general. It is not intended to be a deep analysis of what is in itself a specialised subject. If more detailed information is needed, it can be obtained from the many textbooks on valve design or from the publications produced by various valve manufacturers. (See further reading section of this book for information on three such books.)

A control valve consists of many components which may conveniently be considered as falling into one of two groups: the valve body and the actuator. The former is the part through which the water flows and this flow is controlled by adjusting the resistance offered to the water. This is done by moving the position of a plug in relation to its seat. The position of the plug is controlled by an actuator which acts via the stem.

Figure 6.9 shows a small-bore feed-water control valve body with a contoured trim (the 'trim' being the part of the valve which is in flowing contact with the water). The contour determines the relationship between the position of the plug and the flow of water past it. The type of trim will be dictated by the application, such as the need to minimise acoustic noise or cavitation, the rangeability needed etc. In addition the trim design will determine the valve *characteristic*, which is the curve relating the stem position to the rate of flow of water through the valve. This is an important feature, since the characteristic determines the gain of the valve system, which forms part of the overall loop gain.

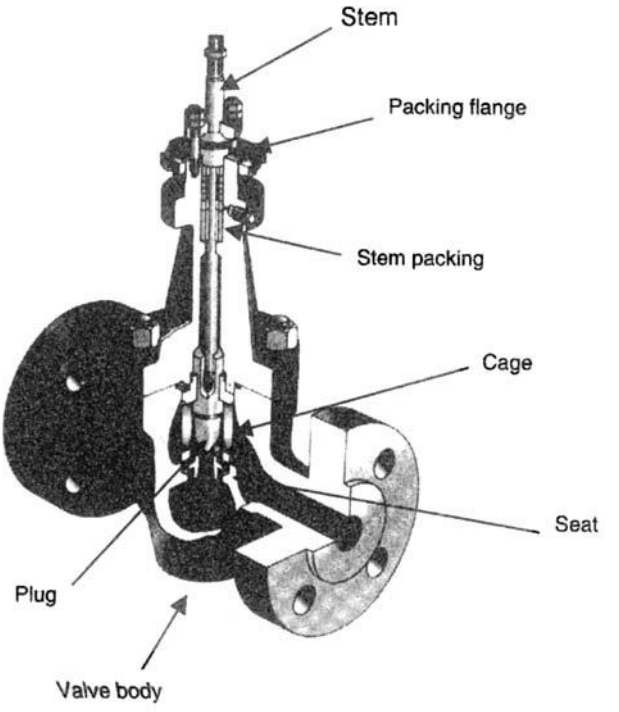


Figure 6.9 *A typical feed-water control valve body*
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For a given opening, the flow through the valve will be determined by the delivery pressure of the feed pump and the resistance that the boiler pipework offers to the flow. To simplify the task of selecting the correct valve size and characteristic, it is necessary to relate everything to a definable set of conditions. This is achieved by determining what the flow through the valve would be if a fixed differential pressure were to be maintained across it. This is termed the *inherent characteristic* of the valve.

Once the valve is operating on the actual plant, the position/flow relationship achieved in practice will not match the inherent characteristic, because in the real world the inlet pressure and system resistance will vary, producing a pressure drop which is different from the value that was used to define the inherent characteristic. The pressure/flow relationship achieved in actual operation is called the *installed characteristic.*

As stated earlier, the gain of the valve is initially defined by the inherent characteristic, three types of which are commonly available, as shown in Figure 6.10. The operation of these different characteristics is now examined.

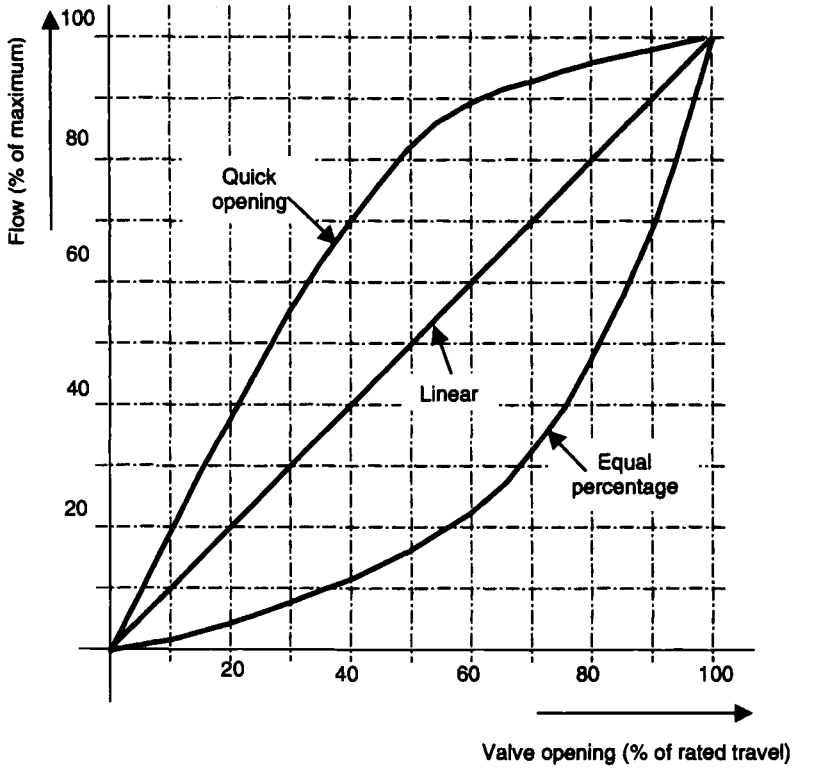


Figure 6.10 Inherent characteristics of valves

6.4.2 *Quick-opening*

With a quick-opening valve, the flow rate through the valve changes very rapidly at low openings, with a slope that is fairly linear. Once the valve has opened about half way the rate of change of flow diminishes. This type of characteristic is usually applied only to shut-off valves.

6.4.3 *Linear*

When a valve has the linear characteristic, the flow rate through it at any given opening (in terms of percentage of maximum flow) is directly equal to the valve stem position (as a percentage of its full travel). With this type of characteristic, the gain of the valve system is constant for all openings. However, the flow through the valve at any given opening depends on the pressure-drop across it and the linear characteristic applies *only if the pressure drop across the valve is constant for each opening*, a condition that may not exist in practice unless special attention is paid to achieving it.

As a general rule, a linear characteristic is preferred for feed-water control applications, since it simplifies tuning of the loop and enables good performance to be achieved over the widest possible range of flows.

Where optimum performance and efficiency is required the additional cost of providing a constant pressure drop across the valve should be considered. Such a solution will add cost, but this will be offset by improved control performance and plant life and by the savings achieved by not running the feed pump or pumps at a higher speed than necessary at reduced loads.

6.4.4 Equal percentage

The third characteristic is called equal percentage. Here, for all positions of the stem, the flow change achieved by moving the stem by a given amount is a constant proportion of the previous flow. What this means is that a given stem movement will change the flow by the same ratio *of the previous flow*, at any point in the valve travel. Therefore, the larger the opening, the greater will be the change of flow produced by a given stem movement.

This is shown by the curve in Figure 6.10, where moving the stem from 20% of full travel to 30% of full travel changes the flow from 5% to 7.5%, while moving the stem from 80% of full travel to 90% of full travel increases the flow from 50% to 75%. In both cases, a stem movement of 10% of full travel results in a 50% increase of the previous flow.

From this it is apparent that, with this type of characteristic, the change of flow for a given stem movement is smallest at low openings and greatest at high openings. In other words, the gain is lower at low flows than it is at high flows. Equal-percentage valves are used where the mechanical plant design is such that there is only a small pressure drop available for the valve, or where the pressure drop across the valve is likely to fluctuate over a wide range.

6.4.5 The valve sizing coefficient

Feed valves are designed to pass a flow that corresponds to the maximum flow requirement of the boiler plus a safety margin, and the capacity of the valve is related to a factor known as the valve sizing coefficient. The factor used widely across the world is based on US units and relates the capacity of the valve to the pressure drop by the following formula:

$$Q = Cv\sqrt{(\Delta P/G)}$$

where

Cv = sizing coefficient
Q = capacity of valve in US gallons per minute (1 US gallon = 3.785 litres)
ΔP = pressure differential across valve in pounds per square inch (psi). (1 psi = 0.069 bar)
G = specific gravity of the flowing fluid

The sizing coefficient is determined by experiment for each type and size of valve using water as the test fluid. It is equal to the volume of water (in US gallons) that will flow through the wide-open valve in one minute when the pressure drop across the valve is 1 psi. The European equivalent of the sizing coefficient is known as *Kv*. To convert between these units multiply the *Kv* figure by 1.66 to obtain the *Cv*.

Given the pressure, temperature, flow and line-size characteristics for a given feed-water application, valve manufacturers will be able to provide detailed guidance on the correct valve size for a given application and from this will be able to predict the pressure drop across the valve.

6.4.6 Fail-safe operation

In the course of designing a feed-water control system, another matter that must be considered is that of selecting the 'fail-safe' position of the plug, the state that will arise if the actuator fails, or if the command signal to the valve is lost. The actual selection will depend on a range of factors, but in determining the safest option it is important to consider the effects of the flow on the valve itself, since the applied pressure may tend to force the plug open or closed. Once again, valve manufacturers will be able to provide advice on this matter in relation to the actual installation being designed.

6.4.7 Selecting the valve size

The size of valve to use in a given application will be determined by many factors, only one of which is the physical size of the line in which it is fitted. Clearly, the valve must be large enough to pass the required flow with ease. Oversized valves will be unnecessarily expensive and should be avoided as unable to effectively control small flows.

For a valve to maintain any control over the process there must be some pressure drop across it, but if the pressure drop is too great a number of undesirable events start to occur.

Figure 6.11 shows in profile the pressure along a short section of pipe containing a restriction (such as a valve). Two curves are shown, one for a valve where careful design produces streamlined flow and minimal pressure loss, and another for a valve through which the flow is more,

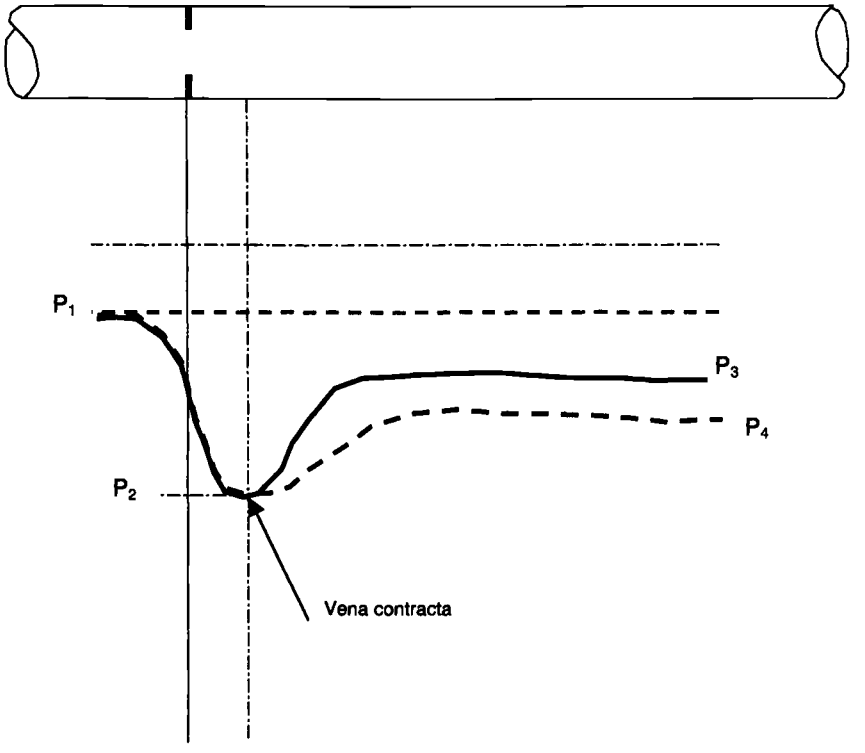


Figure 6.11 Pressure profile along a pipe containing a restriction

producing a greater pressure loss. In each case, the lowest pressure occurs a short distance downstream from the restriction (the Vena contracta) and the pressure rises past this point although it obviously never regains the initial value. If the pressure at the inlet (P_1) is such that the pressure along this curve drops below the vapour pressure, the fluid will start to boil and bubbles will form in it at the point where this occurs.

One of two things will occur as the pressure rises again after the restriction. If it rises above the vapour pressure, the bubbles will collapse—a process known as cavitation, which is accompanied by acoustic noise, the degree of which depends on the scale of the cavitation. Apart from being undesirable on environmental grounds, this noise represents an expenditure of energy. Furthermore, if the bubbles collapse close to metal surfaces, the localised energy release can damage the metal. In some cases this will become apparent as severe pitting of the valve plug and cage, or it could appear as pitting of the pipe itself if the point where the vapour pressure threshold is passed occurs some way down from the valve. High recovery

valves are more likely to experience cavitation, because the downstream pressure is more likely to exceed the vapour pressure.

If, on the other hand, the pressure does not rise above the vapour pressure the bubbles will remain suspended in the fluid, to be carried downstream of the restriction. This is known as 'flashing', and it can result in erosion damage to the valve internals at the point of maximum velocity (usually at or near the point where the plug seats against the ring).

These bubbles reduce the valve's ability to pass fluid and eventually it 'chokes'. When this occurs no more flow can occur irrespective of how much water pressure is applied at the valve inlet. From these considerations it is apparent that the production of bubbles may cause noise and damage to the valve, and possibly the pipework, and it will be correctly surmised that an optimum set of conditions will exist for a given design of valve. In other words, the valve will work best at one pressure-drop point.

To summarise, if the supply of water to a boiler is controlled by throttling the flow through a valve, this can cause erosion and noise and although control can only be effected by maintaining some pressure drop across the valve, this loss represents a loss of energy which should be reduced to the minimum.

6.5 Pumps

The water flowing through the feed valve into the boiler is delivered at pressure by one or more feed pumps. These produce a head of water which is related to the flow through the pump by a characteristic that will be similar to Figure 6.12, which shows that although the discharge pressure remains relatively constant as the flow rises from zero to (in this example) about 50%, beyond this value the pressure tends to decay as the flow increases. From this it will be apparent that the feed valve has to produce a greater pressure drop at low loads than at high loads.

Figure 6.13 shows the flow/pressure characteristic of a pump delivering water into a fixed system. In practice, as the flow through the system increases the resistance offered to it also increases, as shown by the dotted line in the diagram.

At any flow, in order to deliver water into the boiler, the pressure drop across the control valve will be the difference between the pump delivery pressure and the system resistance. For much of the flow range the pressure drop will inevitably be greater or less than the optimum for the valve design. This can be overcome by changing the inlet pressure so that the

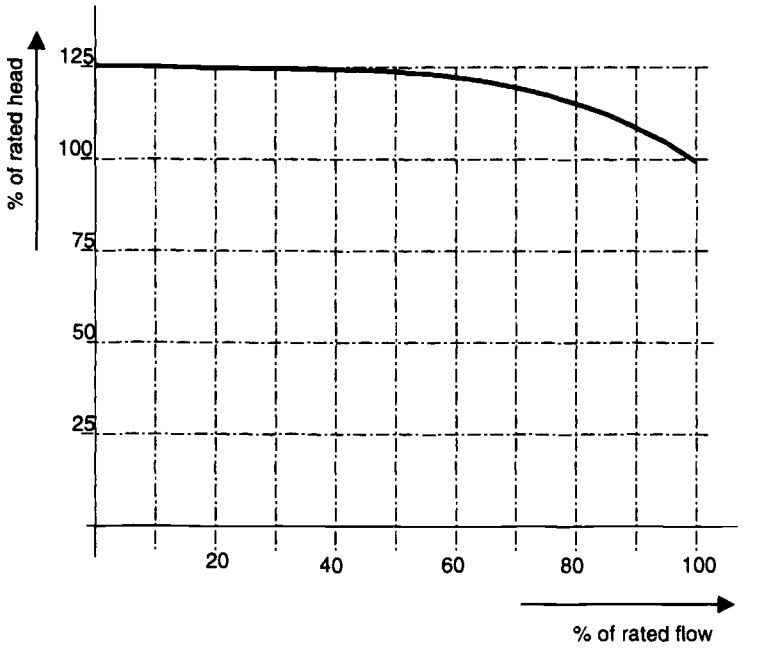


Figure 6.12 Typical speed curve for a feed pump

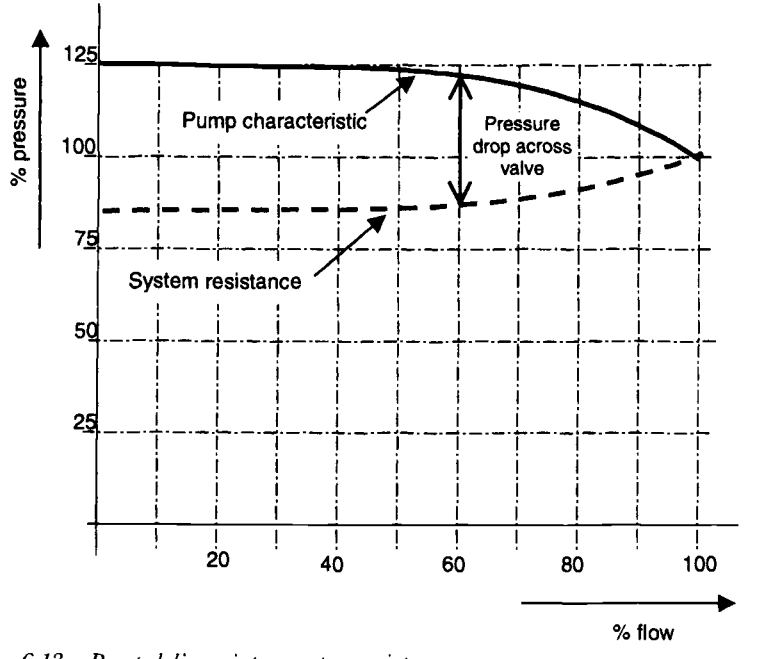


Figure 6.13 Pump delivery into a system resistance

pressure drop always remains at the optimum value. This is achieved by the use of variable-speed pumps.

6.5.1 Variable-speed pumps

Although in many cases the feed pumps operate at a fixed speed, at the design stage consideration should be given to the option of using variable-speed pumps (sometimes known as 'controllable-speed' pumps) because these will enable the feed valve to operate at the optimum pressure conditions for all loads.

The characteristics of a variable-speed pump are shown in Figure 6.14. The four curves show the pressure/flow characteristics at four different operating speeds (A–D), with A being the slowest and D the fastest speed. From this set of curves it will be appreciated that by adjusting the speed of the pump it will be possible to maintain a fixed differential pressure across the feed regulating valve at all boiler loads. In the example shown the pressure drop across the valve (δP) achieved by operating the pump at speed A at 70% flow is maintained at the same value by operating the pump at speed B at 80% flow, speed C at 90% flow etc., as indicated by the lines with arrows at each end.

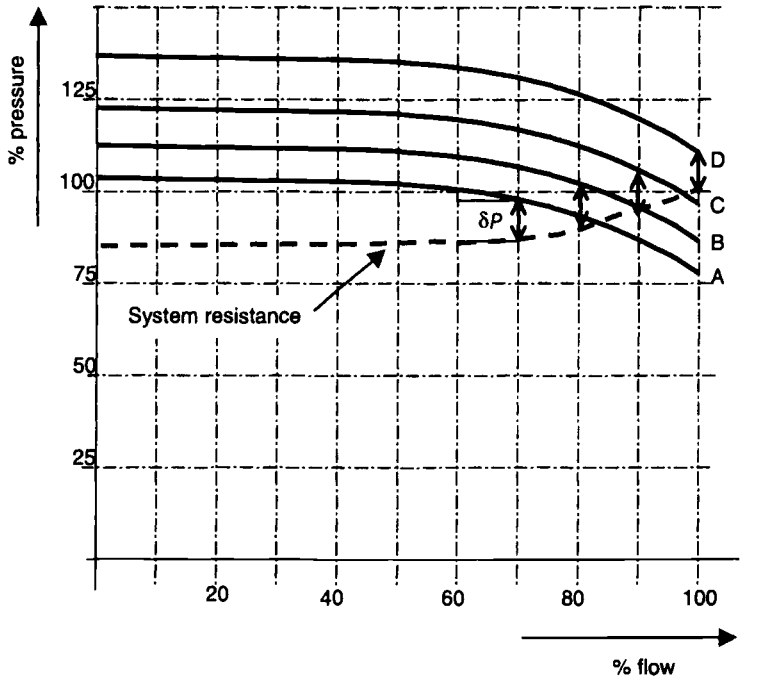


Figure 6.14 Variable-speed pump operation

The advantages of using a variable-speed pump include:

- Improvement of efficiency because of reduced pressure loss.
- Reduction of pumping power.
- Reduction of feed-valve wear due to erosion when operating at low flows.
- Improved control because the valve operates at its designed pressure-drop.
- Improved control offered by the ability to operate with constant loop gain.

Variable-speed pumps are more expensive than fixed-speed ones, but the increase in capital cost is offset by the revenue savings that will be gained, particularly if the boiler operates at reduced throughputs for a significant time over its life. Careful financial analysis will determine whether the savings do justify the additional capital cost, but these calculations must be based on assumptions about the operational regime, and these may change under the influence of external factors which are difficult to predict at the design stage.

Of course, the pumps, valves and boiler pipework are only part of the overall system. Where spray attemperators are used (see Chapter 7), the feed pumps must also be capable of delivering cool water to the nozzles. The design of attemperators requires the water to be at delivered at a pressure which exceeds the steam pressure by a minimum value. Bypassing the pressure drop across the system as shown in Figure 6.15 allows sufficient differential pressure to be maintained at the spray nozzle. But, if a variable-speed pump is used and run down to a low delivery pressure, there is a risk that the required differential may not be available. In such situations, the decision to use fixed- or variable-speed pumps will be affected by the need to maintain an adequate water pressure at the attemporators.

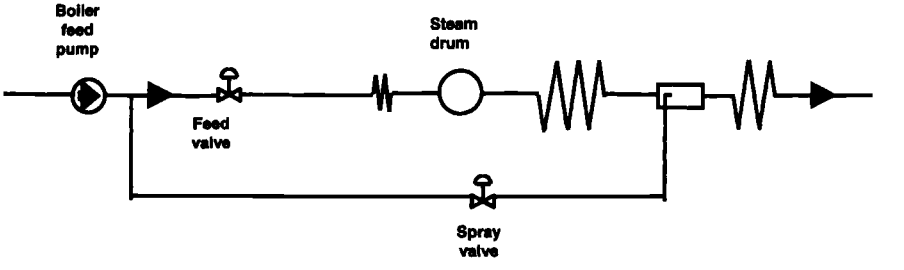


Figure 6.15 Maintaining adequate spray-water differential pressure

6.6 Deaerator control

Strictly speaking, control of the deaerator is not a function of feed-water control which is the subject of this chapter. However, as the deaerator is an essential link in the feed-water supply system it is appropriate to consider its control systems here.

In Figure 2.5 we saw how steam admitted to the deaerator rises upwards past metal trays over which the water is simultaneously cascading downwards. As the water and steam mix and become agitated, entrained gases are released. The dissolved gases are vented to the atmosphere because the vessel is pressurised by the steam. The deaerator is situated in the water circuit between the discharge of the condenser extraction pump and the inlet of the feed pumps, as shown in Figure 6.16.

It will be evident that two control functions are required by the deaerator: one to maintain the steam pressure at the optimum value, the other to keep the storage vessel roughly half full of water.

6.6.1 Steam pressure control

The pressure of the steam entering the deaerator is maintained by a simple controller whose measured-value signal is obtained from a trans-

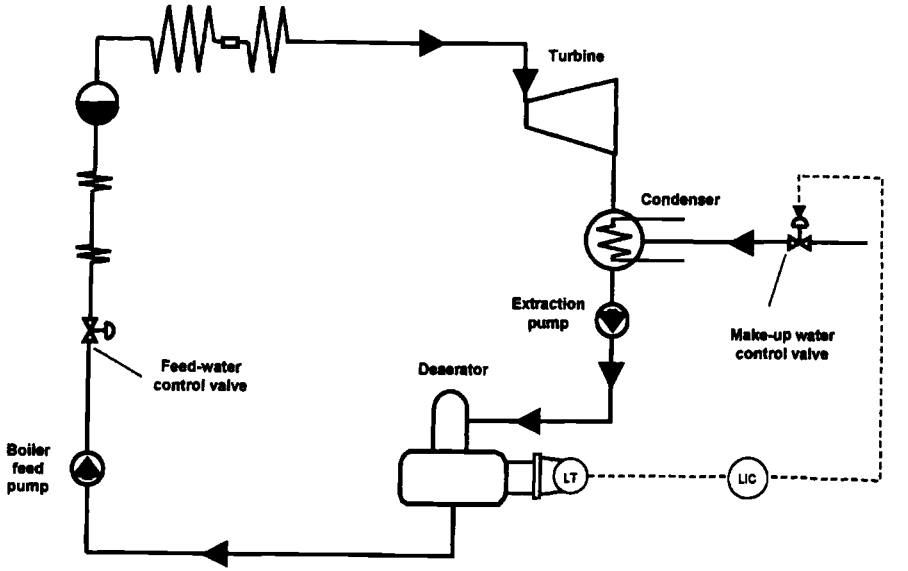


Figure 6.16 Principle of deaerator level control system

mitter measuring the steam pressure in the deaerator. The set value of the controller is normally fixed.

It has already been explained that the steam supply may be obtained either from the boiler or from an extraction point on the turbine. If the latter source is used, special consideration has to be given to ensuring that an event such as a turbine trip does not deprive the deaerator of the steam supply it needs for its operation. This purpose is served by taking a steam supply from the boiler and passing it to the deaerator via a pressure-reducing and desuperheating system (PRDS). This steam supply is referred to as 'pegging steam'. Judicious adjustment of the PRDS controller setpoint will ensure that an adequate steam supply for the deaerator is obtained at all times. However, to ensure rapid response to a turbine trip, a system of interlocks should be provided, so that the pegging steam supply is brought into service immediately on detection of a trip.

6.6.2 Level control

The storage vessel provides a measure of reserve capacity for the plant. To achieve this function the level of water in it must be maintained at roughly the midpoint. This is achieved by means of a level controller whose measured-value signal is obtained from a differential-pressure transmitter or from capacitive probes which would normally be connected to tappings of an external water column which is in turn connected to the top and bottom of the deaerator storage vessel. If there were no losses in the system, the amount of water would be constant and the level in the deaerator storage vessel would remain at the correct value set during commissioning. However, losses are inevitable (for example, due to leakages at pump glands or during soot-blowing or blowdown operations), and a supply of treated water must therefore be made available. The deaerator level controller output adjusts the opening of a valve that admits this make-up water to the condenser, as shown in Figure 6.16.

The make-up supply is conventionally fed into the system at the condenser. Figure 6.16 shows that interaction between the level controllers of the deaerator and condenser is inevitable. The situation is made more complex because the condenser extraction pump has to be provided with a bypass arrangement to maintain a minimum flow through the pump at all times.

In fact, the conditions which cause the deaerator level controller to call for more water to be added to the system will also cause the condenser level to fall, and so the two systems do not act in opposite senses. Nevertheless they do interact, and care must be taken to minimise the instability that is likely to arise.

6.6.2.1 Spill control

In addition to the level control system described above, a system must be provided to drain excess water from the deaerator storage vessel in the event of an oversupply of water and steam. This function is normally achieved on large power-station boilers by using the output of the level controller to operate the make-up and spill valves in split range. A typical arrangement is shown in Table 6.1. Smaller boilers will be limited to having a spill valve whose opening is normally controlled on an on/off basis from a high-level alarm.

6.6.3 Integrated level control

The long dead time and long time constant of the deaerator level system complicates the design and commissioning of this control loop. This situation has been worsened by the commercial pressure to operate power-station boilers at partial load and the increased need for two-shifting operations. (The term 'two-shifting' refers to operations where the plant runs for two eight-hour periods (or shifts) in a working day, and is shut down for the remaining eight-hour shift.) There is also a drive to reduce the consumption of controllable make-up water, although this necessarily increases with the amount of load changing to which the boiler is subjected. A typical scenario is that if the boiler experiences a large load reduction (say 50%) the deaerator level would tend to rise because as the steam flow from the boiler reduces so must the feed flow. During this time, the condensate flowing to the deaerator would not be required by the feed pump, causing the level in the storage vessel to rise. The level controls would react by dumping the excess water that is filling the deaerator. If there is a load increase to the original load, the deaerator level would drop due to the increased flow from the feed pump, and the level controls would tend to add make-up water to the condenser. This cycle of making-up and spilling can occur every time the load changes.

Table 6.1 Operation of deaerator level-control valves
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Controller output	Valve duty	Valve position
0–25%	Spill valve	100–0%
25–75%	Deadband	n/a
75–100%	Make-up valve	0–100%

The requirement to reduce the wasted make-up water has resulted in the use of advanced control strategies for controlling the make-up and spill on larger boilers. For example a fuzzy logic control algorithm embedded in the DCS software can be used. The inputs to the controller are the steam flow and the deaerator level, the outputs are to the make-up and spill valves. The system uses rule-based logic to take decisions on making-up and spilling. The goal of the scheme is to minimise make-up and spill by retaining water within the condensate system until it is required.

6.7 Summary

Once the combustion process has occurred, the feed water has boiled and the steam has been generated, the next requirement is to ensure that the temperature of the steam that is delivered to the turbine or heat load is maintained at the correct value. In the next chapter we shall look at the control and instrumentation systems that are employed for this purpose.